

Ahmed Abd El-Azeem

AI Engineer | LLM Integration Specialist

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-grade LLM applications, agentic workflows, retrieval-augmented generation pipelines, and AI-backed business systems. Strong background in Python backend development with FastAPI, data modeling, document processing, semantic retrieval, and API delivery. Experienced in creating domain-specific solutions for AEC and MEP workflows, with practical knowledge of prompt engineering, vector databases.

CORE SKILLS

Languages	Python, SQL, C#
Frameworks	FastAPI, LangChain, LangGraph, Pydantic, SQLAlchemy, Alembic, Streamlit, .NET Core
AI & Data	Google Gemini API, Prompt Engineering, ChromaDB, Sentence Transformers, RAG, Document Ingestion, Semantic Search

PROFESSIONAL EXPERIENCE

PROJECTS

IFC AI Assistant — BIM Agentic System | *Python · LangChain · LangGraph · Google Gemini API · Streamlit · Pydantic*

- Built a domain-specific AI agent for querying, analyzing, and authoring IFC/BIM files with 20+ tools covering geometry analysis, clash detection, material inspection, and model validation.
- Applied a Plan → Execute → Report workflow to ensure outputs remained grounded in tool usage and model data rather than unsupported generation.

Data Brain Portal — MEP Materials Intelligence Platform | *Python · FastAPI · LangChain · Google Gemini API · ChromaDB · Sentence Transformers*

- Built an AI-powered materials intelligence platform that centralizes vendor datasheets and makes product information semantically searchable for engineering teams.
- Developed ingestion pipelines for PDFs and images of vendor catalogues, extracted structured metadata, and indexed embeddings for grounded search and analysis.
- Implemented BOQ analysis and exportable Excel reporting workflows for engineering use cases.

BOQ Pricer Platform — AI-Powered Construction Pricing Backend | *Python · FastAPI · SQLAlchemy · PostgreSQL · Alembic · LangChain · Google Gemini API · ChromaDB · Sentence Transformers*

- Built an AI-driven pricing backend with authentication, role-based access, project management APIs, and persistent job storage.
- Developed semantic retrieval pipelines to parse multi-sheet BOQ files, extract cost elements, and support project-scoped similarity search and pricing assistance.

TECHNICAL SKILLS

- Object Oriented Programming (OOP)
- Data Structure
- Data analysis
- AI LLM integration

EDUCATION

B.Sc. in Civil Engineering | Helwan University | 2018

CERTIFICATIONS & ADDITIONAL INFORMATION

- Languages: Arabic (Native), English (Very Good), Russian (Beginner).
- AWS Data Engineer Specialization
- Machine learning Specialization with Anderw Ng.